Discussion 7: Convexity of Functions and Sets

1. Background: What is Convexity?

Convexity is one of the most important ideas in optimization. If a problem is "convex," it becomes much easier to solve. This section builds everything from scratch.

1.1 Convex Sets (the simpler idea; start here)

Definition 1 (Convex Set)**.** A set $C \subseteq \mathbb{R}^n$ is **convex** if for every two points in the set, the entire straight line between them also stays inside the set.

Formally: for all $\vec{x}_1, \vec{x}_2 \in C$ and all $\theta \in [0, 1]$:

$$\theta\vec{x}_1 + (1-\theta)\vec{x}_2 \in C$$

Let's unpack the formula. The expression $\theta\vec{x}_1 + (1-\theta)\vec{x}_2$ is called a **convex combination**. As θ goes from 0 to 1, this expression traces out the line segment from $\vec{x}_2$ to $\vec{x}_1$:

- When $\theta = 0$: the expression gives $0 \cdot \vec{x}_1 + 1 \cdot \vec{x}_2 = \vec{x}_2$ (the starting point).
- When $\theta = \frac{1}{2}$: the expression gives $\frac{1}{2}\vec{x}_1 + \frac{1}{2}\vec{x}_2$ (the midpoint).
- When $\theta = 1$: the expression gives $1 \cdot \vec{x}_1 + 0 \cdot \vec{x}_2 = \vec{x}_1$ (the ending point).

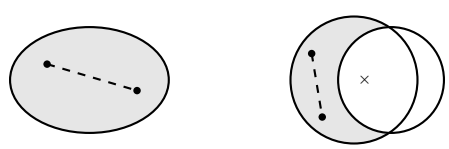
So the definition says: pick any two points in C, and every point on the line between them must also be in C.

Example 1 (Concrete numbers)**.** Let $\vec{x}_1 = (2, 4)$ and $\vec{x}_2 = (6, 0)$. The point at $\theta = 0.3$ is:

$$0.3 \cdot (2, 4) + 0.7 \cdot (6, 0) = (0.6 + 4.2,\ 1.2 + 0) = (4.8,\ 1.2)$$

For the set to be convex, $(4.8, 1.2)$ must be in the set.

Example 2 (Convex vs. not convex)**.**



Convex Not convex

- **Left (convex):** An ellipse. Any line between two interior points stays inside.
- **Right (not convex):** A crescent. The dashed line passes through the "bite" taken out of the shape, leaving the set.

Common convex sets:

- A filled circle or ball: convex.
- A rectangle or box: convex.
- A halfspace $\{\vec{x} : \vec{a}^\top \vec{x} \leq b\}$: convex. (Everything on one side of a flat boundary.)
- A single point: convex. A line: convex. All of $\mathbb{R}^n$: convex. The empty set: convex.

Common non-convex sets:

- A donut (annulus): not convex.
- The outside of a circle $\{(x, y) : x^2 + y^2 \geq 1\}$: not convex.

1.2 Convex Functions

Definition 2 (Convex Function)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is **convex** if two conditions hold:

1. Its domain $\mathrm{dom}(f)$ is a convex set, and

2. For all $\vec{x}, \vec{y} \in \mathrm{dom}(f)$ and $\theta \in [0, 1]$:

$$f\big(\theta\vec{x} + (1-\theta)\vec{y}\big) \leq \theta f(\vec{x}) + (1-\theta) f(\vec{y})$$

Let's understand each side of this inequality.

- **Left side:** $f\big(\theta\vec{x} + (1-\theta)\vec{y}\big)$. This is: "first blend the inputs, then evaluate f." We take a point on the line between $\vec{x}$ and $\vec{y}$, and ask what is f there?
- **Right side:** $\theta f(\vec{x}) + (1-\theta) f(\vec{y})$. This is: "first evaluate f at each endpoint, then blend the outputs." We compute f at $\vec{x}$ and at $\vec{y}$, then take the same weighted average of those values.

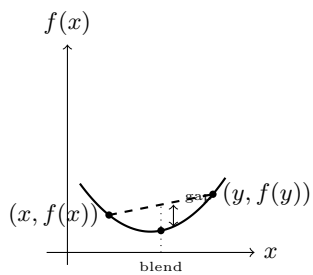
The inequality says: the function value at the blended input $\leq$ the blend of the function values. In other words, the function lies *below* (or on) the line segment connecting the two points on its graph.

Example 3 (Verifying $f(x) = x^2$ is convex with actual numbers)**.** Let $x = 1$, $y = 3$, $\theta = 0.5$ (midpoint). Check:

$$\textbf{Left side: } f(0.5 \cdot 1 + 0.5 \cdot 3) = f(2) = 4$$
$$\textbf{Right side: } 0.5 \cdot f(1) + 0.5 \cdot f(3) = 0.5 \cdot 1 + 0.5 \cdot 9 = 5$$

Indeed $4 \leq 5$. ✓ The function at the midpoint ($= 4$) is below the average of the endpoint values ($= 5$).

Visual picture



The solid curve is f. The dashed line connects two points on f. Convexity means the curve always stays below (or on) the dashed line. The "gap" is the difference: right side minus left side ≥ 0.

Familiar convex functions:

- $f(x) = x^2$: parabola curving up.
- $f(x) = |x|$: V-shape.
- $f(x) = e^x$: exponential.
- $f(\vec{x}) = \vec{a}^\top \vec{x} + b$: any linear/affine function is both convex and concave.

Familiar concave functions (curve downward):

- $f(x) = \log(x)$ for $x > 0$.
- $f(x) = \sqrt{x}$ for $x \geq 0$.
- $f(x) = -x^2$: upside-down parabola.

1.3 Concave Functions

Definition 3 (Concave Function)**.** f is **concave** if $-f$ is convex. Equivalently, the inequality flips:

$$f(\theta\vec{x} + (1-\theta)\vec{y}) \geq \theta f(\vec{x}) + (1-\theta) f(\vec{y})$$

The function lies *above* the line segment. A function that curves downward (like log or $\sqrt{x}$) is concave.

Special case: Any affine function $f(\vec{x}) = \vec{a}^\top \vec{x} + b$ satisfies both $\leq$ and $\geq$ with equality, so it is both convex and concave.

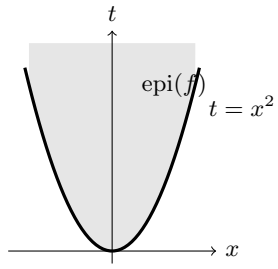
1.4 The Epigraph

Definition 4 (Epigraph)**.** The **epigraph** of f is the set of all points "on or above" the graph:

$$\text{epi}(f) = \{(\vec{x}, t) : \vec{x} \in \text{dom}(f),\ t \geq f(\vec{x})\}$$

Think of it this way: plot f in $n+1$ dimensions (the input plus one extra axis for the output). Now shade everything above the surface. That shaded region is the epigraph.

Example 4 (Epigraph of $f(x) = x^2$)**.**



The shaded region (above the parabola) is the epigraph. It is a convex set, which reflects the fact that x^2 is a convex function.

Key connection: f is convex $\iff$ epi(f) is a convex set.
This is powerful because it lets us translate between function convexity and set convexity.

1.5 Three Tests for Convexity

Test 1: Definition (works for any function, even non-differentiable). Check the inequality directly:

$$f(\theta\vec{x} + (1-\theta)\vec{y}) \leq \theta f(\vec{x}) + (1-\theta) f(\vec{y})$$

Test 2: First-order condition (requires f differentiable). f is convex $\iff$ for all $\vec{x}, \vec{y}$:

$$f(\vec{y}) \geq f(\vec{x}) + \nabla f(\vec{x})^\top (\vec{y} - \vec{x})$$

What $\nabla f(\vec{x})$ is: The gradient of f at $\vec{x}$, a vector of partial derivatives. The right side is the tangent line (or tangent plane) at $\vec{x}$. The condition says: a convex function always lies above all its tangent lines.

Test 3: Second-order condition (requires f twice differentiable). f is convex $\iff$ the Hessian matrix $\nabla^2 f(\vec{x})$ is positive semidefinite (PSD) everywhere.
What the Hessian is: The $n \times n$ matrix of all second partial derivatives. Positive semidefinite means all eigenvalues ≥ 0.
For single-variable functions: This simplifies to $f''(x) \geq 0$ for all x. (The second derivative is nonnegative everywhere.)

Example 5 (Checking $f(x) = x^2$ with all three tests)**.**
Test 1: $(\theta a + (1-\theta)b)^2 \leq \theta a^2 + (1-\theta)b^2$? Yes, this is equivalent to $\theta(1-\theta)(a-b)^2 \geq 0$, which is always true.
Test 2: $f'(x) = 2x$. Check: $y^2 \geq x^2 + 2x(y-x) = 2xy - x^2$. This simplifies to $(y-x)^2 \geq 0$. True. ✓
Test 3: $f''(x) = 2 > 0$. Positive everywhere. ✓

1.6 Jensen's Inequality

Jensen's Inequality: If f is convex and $\theta_1 + \theta_2 + \cdots + \theta_k = 1$ with all $\theta_i \geq 0$:

$$f(\theta_1\vec{x}_1 + \cdots + \theta_k\vec{x}_k) \leq \theta_1 f(\vec{x}_1) + \cdots + \theta_k f(\vec{x}_k)$$

This is just the convexity definition extended from 2 points to k points. It says: "the function at the weighted average is at most the weighted average of the function values."

2. Problem 1(a): Restriction to a Line

Problem: Show that f is convex if and only if for all $\vec{x} \in \mathrm{dom}(f)$ and all $\vec{v}$, the function $g(t) = f(\vec{x} + t\vec{v})$ is convex.

2.1 What is $g(t) = f(\vec{x} + t\vec{v})$?

This is a way to turn a multivariate function into a single-variable function by "slicing" it along a line.

Building intuition step by step

Imagine f is a function on $\mathbb{R}^2$ (like a hilly landscape).
Step 1: Pick a starting point $\vec{x}$. This is a specific location on the landscape.
Step 2: Pick a direction $\vec{v}$. This is the direction you will walk.
Step 3: As you vary t, the point $\vec{x} + t\vec{v}$ moves along a straight line through $\vec{x}$ in direction $\vec{v}$. When $t = 0$, you are at $\vec{x}$. When $t > 0$, you move forward. When $t < 0$, you move backward.
Step 4: $g(t) = f(\vec{x} + t\vec{v})$ records the "altitude" as you walk. It is a 1D function of the single variable t.
The claim: f (the whole landscape) is convex if and only if g (the altitude profile along every possible line) is convex.

Example 6 (With actual numbers)**.** Let $f(x_1, x_2) = x_1^2 + x_2^2$. Pick $\vec{x} = (1, 0)$ and $\vec{v} = (1, 2)$.

Then $\vec{x} + t\vec{v} = (1 + t,\ 2t)$, so:

$$g(t) = f(1 + t,\ 2t) = (1 + t)^2 + (2t)^2 = 1 + 2t + t^2 + 4t^2 = 5t^2 + 2t + 1$$

This is a quadratic in t with positive leading coefficient ($5 > 0$), so it is a convex function of t. ✓

2.2 Proof: ($\Rightarrow$) f convex implies g convex

Goal: Assume f is convex. Show $g(t) = f(\vec{x} + t\vec{v})$ is convex.

What we need to show: For any t_1, t_2 and $\lambda \in [0, 1]$:

$$g(\lambda t_1 + (1 - \lambda)t_2) \leq \lambda g(t_1) + (1 - \lambda)g(t_2)$$

Step 1: Write out the left side using the definition of g.

$$g(\lambda t_1 + (1 - \lambda)t_2) = f\big(\vec{x} + (\lambda t_1 + (1 - \lambda)t_2)\vec{v}\big)$$

Step 2: Rearrange the argument of f into a convex combination.

This is the key algebraic manipulation. Distribute $\vec{v}$:

$$\vec{x} + (\lambda t_1 + (1 - \lambda)t_2)\vec{v} = \vec{x} + \lambda t_1\vec{v} + (1 - \lambda)t_2\vec{v}$$

Now we use a trick: since $\lambda + (1 - \lambda) = 1$, we can write $\vec{x} = \lambda\vec{x} + (1 - \lambda)\vec{x}$:

$$\begin{aligned} &= \lambda\vec{x} + (1 - \lambda)\vec{x} + \lambda t_1\vec{v} + (1 - \lambda)t_2\vec{v} \\ &= \lambda(\vec{x} + t_1\vec{v}) + (1 - \lambda)(\vec{x} + t_2\vec{v}) \end{aligned}$$

This is a convex combination of $(\vec{x} + t_1\vec{v})$ and $(\vec{x} + t_2\vec{v})$.

Step 3: Apply convexity of f.

Since f is convex and we have a convex combination:

$$\begin{aligned} f\big(\lambda(\vec{x} + t_1\vec{v}) + (1 - \lambda)(\vec{x} + t_2\vec{v})\big) &\leq \lambda f(\vec{x} + t_1\vec{v}) + (1 - \lambda)f(\vec{x} + t_2\vec{v}) \\ &= \lambda g(t_1) + (1 - \lambda)g(t_2) \end{aligned}$$

Step 4: Combine.

Putting Steps 1–3 together:

$$g(\lambda t_1 + (1 - \lambda)t_2) \leq \lambda g(t_1) + (1 - \lambda)g(t_2)$$

This is exactly the definition of g being convex. □

Step 2 is the only hard part

The entire proof boils down to one algebraic trick: splitting $\vec{x}$ into $\lambda\vec{x} + (1 - \lambda)\vec{x}$ and regrouping. Everything else is just applying definitions. If you remember this trick, you can reproduce the proof.

2.3 Proof: ($\Leftarrow$) Every g convex implies f convex

Goal: Assume $g(t) = f(\vec{x} + t\vec{v})$ is convex for every choice of $\vec{x}$ and $\vec{v}$. Show f is convex.

What we need to show: For any $\vec{x}_1, \vec{x}_2 \in \text{dom}(f)$ and $\lambda \in [0, 1]$:

$$f(\lambda\vec{x}_1 + (1-\lambda)\vec{x}_2) \leq \lambda f(\vec{x}_1) + (1-\lambda)f(\vec{x}_2)$$

Step 1: Cleverly choose $\vec{x}$ and $\vec{v}$ to define g.

Set $\vec{x} = \vec{x}_2$ and $\vec{v} = \vec{x}_1 - \vec{x}_2$. Then define:

$$g(t) = f(\vec{x}_2 + t(\vec{x}_1 - \vec{x}_2))$$

Step 2: Check what g gives at specific values of t.

- $g(0) = f(\vec{x}_2 + 0 \cdot (\vec{x}_1 - \vec{x}_2)) = f(\vec{x}_2)$
- $g(1) = f(\vec{x}_2 + 1 \cdot (\vec{x}_1 - \vec{x}_2)) = f(\vec{x}_1)$
- $g(\lambda) = f(\vec{x}_2 + \lambda(\vec{x}_1 - \vec{x}_2)) = f(\lambda\vec{x}_1 + (1-\lambda)\vec{x}_2)$

Note: $\vec{x}_2 + \lambda(\vec{x}_1 - \vec{x}_2) = \vec{x}_2 + \lambda\vec{x}_1 - \lambda\vec{x}_2 = \lambda\vec{x}_1 + (1-\lambda)\vec{x}_2$. This is just algebra.

Step 3: Apply convexity of g.

g is convex by assumption. Apply the definition with inputs $t_1 = 1$, $t_2 = 0$, and weight λ:

$$g(\lambda \cdot 1 + (1-\lambda) \cdot 0) \leq \lambda \cdot g(1) + (1-\lambda) \cdot g(0)$$

Step 4: Substitute the values from Step 2.

The left side is $g(\lambda) = f(\lambda\vec{x}_1 + (1-\lambda)\vec{x}_2)$.

The right side is $\lambda g(1) + (1-\lambda)g(0) = \lambda f(\vec{x}_1) + (1-\lambda)f(\vec{x}_2)$.

So:

$$f(\lambda\vec{x}_1 + (1-\lambda)\vec{x}_2) \leq \lambda f(\vec{x}_1) + (1-\lambda)f(\vec{x}_2)$$

This is exactly the definition of f being convex. $\square$

Intuition: Why this result is useful

Checking if a function on $\mathbb{R}^{100}$ is convex sounds hard. But this result says: just check if every 1D slice is convex. And for a 1D function, convexity is easy to check (just verify $g''(t) \geq 0$).

3. Problem 1(b): Pointwise Maximum

Problem: If f_1 and f_2 are convex, show that $f(\vec{x}) = \max(f_1(\vec{x}), f_2(\vec{x}))$ is convex.

3.1 What is the pointwise maximum?

At each point $\vec{x}$, we evaluate both $f_1(\vec{x})$ and $f_2(\vec{x})$, and take whichever is larger.

Example 7 (Concrete)**.** Let $f_1(x) = 2x + 1$ and $f_2(x) = -x + 4$ (both lines, hence convex).

x	$f_1(x)$	$f_2(x)$	$f(x) = \max$
0	1	4	4
1	3	3	3
2	5	2	5

The graph of f follows f_2 when $x < 1$ and f_1 when $x > 1$, creating a V-shape: still convex.

3.2 Proof 1: Via Epigraphs (most elegant)

The idea: We will show that $\text{epi}(f) = \text{epi}(f_1) \cap \text{epi}(f_2)$, then use the fact that intersections of convex sets are convex.

Step 1: Check the domain.

$\text{dom}(f) = \text{dom}(f_1) \cap \text{dom}(f_2)$. We will prove in Problem 2 that the intersection of convex sets is convex, so $\text{dom}(f)$ is convex. ✓

Step 2: Write out what $\text{epi}(f)$ is.

$$\begin{aligned}\text{epi}(f) &= \{(\vec{x}, t) : f(\vec{x}) \leq t\} \\ &= \{(\vec{x}, t) : \max(f_1(\vec{x}), f_2(\vec{x})) \leq t\}\end{aligned}$$

Step 3: Simplify $\max(a, b) \leq t$.

The inequality $\max(a, b) \leq t$ is true if and only if *both* $a \leq t$ and $b \leq t$. (The larger of the two is at most t, so both must be at most t.)

$$\begin{aligned}\text{epi}(f) &= \{(\vec{x}, t) : f_1(\vec{x}) \leq t \text{ and } f_2(\vec{x}) \leq t\} \\ &= \{(\vec{x}, t) : f_1(\vec{x}) \leq t\} \ \cap \ \{(\vec{x}, t) : f_2(\vec{x}) \leq t\} \\ &= \text{epi}(f_1) \cap \text{epi}(f_2)\end{aligned}$$

Step 4: Conclude.

f_1, f_2 convex $\Rightarrow \text{epi}(f_1), \text{epi}(f_2)$ are convex sets $\Rightarrow \text{epi}(f_1) \cap \text{epi}(f_2)$ is convex (intersection of convex sets) $\Rightarrow \text{epi}(f)$ is convex $\Rightarrow f$ is convex. □

Why this proof is nice

It turns a question about functions into a question about sets. The key fact $\max(a, b) \leq t \iff a \leq t$ and $b \leq t$ converts the pointwise max into an intersection.

3.3 Proof 2: Direct from the definition (most concrete)

Step 1: Set up.

Take $\vec{x}_1, \vec{x}_2 \in \text{dom}(f)$, $\lambda \in [0, 1]$. Let $\vec{z} = \lambda\vec{x}_1 + (1 - \lambda)\vec{x}_2$. We need:

$$f(\vec{z}) \leq \lambda f(\vec{x}_1) + (1 - \lambda) f(\vec{x}_2)$$

Step 2: Case split on which function achieves the max at $\vec{z}$.

Since $f(\vec{z}) = \max(f_1(\vec{z}), f_2(\vec{z}))$, either $f(\vec{z}) = f_1(\vec{z})$ or $f(\vec{z}) = f_2(\vec{z})$.

Case 1: $f(\vec{z}) = f_1(\vec{z})$.

$$\begin{aligned}f(\vec{z}) &= f_1(\lambda\vec{x}_1 + (1 - \lambda)\vec{x}_2) \\ &\leq \lambda f_1(\vec{x}_1) + (1 - \lambda) f_1(\vec{x}_2) && \text{(convexity of } f_1\text{)} \\ &\leq \lambda \underbrace{\max(f_1(\vec{x}_1), f_2(\vec{x}_1))}_{=f(\vec{x}_1)} + (1 - \lambda) \underbrace{\max(f_1(\vec{x}_2), f_2(\vec{x}_2))}_{=f(\vec{x}_2)} && \text{(since } f_i \leq \max\text{)}\end{aligned}$$

The second inequality uses: $f_1(\vec{x}_1) \leq \max(f_1(\vec{x}_1), f_2(\vec{x}_1))$ (a number is at most the max of it and something else). Same for $f_1(\vec{x}_2)$.

Case 2: $f(\vec{z}) = f_2(\vec{z})$. Identical, with f_2 in place of f_1.

In both cases, $f(\vec{z}) \leq \lambda f(\vec{x}_1) + (1 - \lambda) f(\vec{x}_2)$. □

3.4 Proof 3: Using a lemma about max

Step 1: Prove a useful inequality.

Lemma: For any real numbers a, b, c, d:

$$\max(a+b,\ c+d) \leq \max(a,c) + \max(b,d)$$

Proof of lemma: We know $a \leq \max(a,c)$ and $b \leq \max(b,d)$. Adding: $a+b \leq \max(a,c)+\max(b,d)$. Similarly, $c+d \leq \max(a,c)+\max(b,d)$. Since both $a+b$ and $c+d$ are $\leq$ the right side, so is their max. $\square$

Step 2: Chain the inequalities.

$$\begin{aligned}
&f(\lambda\vec{x}_1 + (1-\lambda)\vec{x}_2) \\
&= \max\big(f_1(\lambda\vec{x}_1 + (1-\lambda)\vec{x}_2),\ f_2(\lambda\vec{x}_1 + (1-\lambda)\vec{x}_2)\big) \\
&\leq \max\big(\lambda f_1(\vec{x}_1) + (1-\lambda)f_1(\vec{x}_2),\ \lambda f_2(\vec{x}_1) + (1-\lambda)f_2(\vec{x}_2)\big) \\
&\leq \max\big(\lambda f_1(\vec{x}_1), \lambda f_2(\vec{x}_1)\big) + \max\big((1-\lambda)f_1(\vec{x}_2), (1-\lambda)f_2(\vec{x}_2)\big) \\
&= \lambda \max(f_1(\vec{x}_1), f_2(\vec{x}_1)) + (1-\lambda)\max(f_1(\vec{x}_2), f_2(\vec{x}_2)) \\
&= \lambda f(\vec{x}_1) + (1-\lambda)f(\vec{x}_2) \quad \square
\end{aligned}$$

Explanation of each line:

- Line 1 $\to$ 2: Apply convexity of f_1 and f_2 inside the max. Each f_i at the blend $\leq$ blend of f_i values.
- Line 2 $\to$ 3: Apply the lemma.
- Line 3 $\to$ 4: Factor out λ and $(1-\lambda)$. For example, $\max(\lambda a, \lambda b) = \lambda \max(a,b)$ when $\lambda \geq 0$.

Generalization

This extends to any number of functions: if $f_1, \ldots, f_m$ are all convex, then $f(\vec{x}) = \max_i f_i(\vec{x})$ is convex. This even works for infinitely many functions: $f(\vec{x}) = \sup_\alpha f_\alpha(\vec{x})$ is convex if every f_α is convex.

4. Problem 2(a): Intersection of Convex Sets

Problem: If C_1 and C_2 are convex sets, show $C = C_1 \cap C_2$ is convex.

4.1 What is $C_1 \cap C_2$?

$C_1 \cap C_2$ is the set of points that are in *both* C_1 and C_2 simultaneously: the overlap region.

Example 8. If C_1 is a circle and C_2 is a rectangle (both convex), their intersection is a "rounded rectangle" shape, which is also convex.

4.2 Proof

Step 1: Take two arbitrary points in C.

Let $\vec{x}_1, \vec{x}_2 \in C = C_1 \cap C_2$ and $\theta \in [0,1]$.

Since $\vec{x}_1 \in C_1 \cap C_2$, we know:

- $\vec{x}_1 \in C_1$ and $\vec{x}_1 \in C_2$

Similarly:

- $\vec{x}_2 \in C_1$ and $\vec{x}_2 \in C_2$

Step 2: Apply convexity of C_1.

C_1 is convex, and $\vec{x}_1, \vec{x}_2 \in C_1$, so by definition of convex set:

$$\theta\vec{x}_1 + (1-\theta)\vec{x}_2 \in C_1 \quad \checkmark$$

Step 3: Apply convexity of C_2.

C_2 is convex, and $\vec{x}_1, \vec{x}_2 \in C_2$, so:

$$\theta\vec{x}_1 + (1-\theta)\vec{x}_2 \in C_2 \quad \checkmark$$

Step 4: Conclude.

The point $\theta\vec{x}_1 + (1-\theta)\vec{x}_2$ is in C_1 (by Step 2) and in C_2 (by Step 3). Therefore it is in $C_1 \cap C_2 = C$. $\square$

The logic in one sentence

Each set individually "keeps" the convex combination inside. Since both sets keep it, the intersection keeps it too.

This generalizes: The intersection of *any* number of convex sets (even infinitely many) is convex. The proof is identical: if the point is in every set, convexity of each set keeps the convex combination in that set, so the combination is in the intersection.

Warning: Union does NOT preserve convexity.
Consider $C_1 = \{(x, y) : x \leq -1\}$ and $C_2 = \{(x, y) : x \geq 1\}$. Both are convex (halfspaces). But $C_1 \cup C_2$ is not convex: the point $(-1, 0) \in C_1$ and $(1, 0) \in C_2$, but the midpoint $(0, 0)$ is in neither.

Why this result matters in practice

Feasible sets in optimization are intersections of constraints. Each constraint like $\vec{a}^\top\vec{x} \leq b$ defines a halfspace (convex). A system of constraints defines their intersection, which is convex by this result. This is why linear programs and convex programs have nice feasible sets.

Key Takeaways

1. **Convex set:** Every line between two points stays inside. **Convex function:** The graph lies below every chord.
2. **Restriction to a line:** f is convex if and only if every 1D slice $g(t) = f(\vec{x} + t\vec{v})$ is convex. The proof trick is writing $\vec{x} = \lambda\vec{x} + (1-\lambda)\vec{x}$.
3. **Pointwise max preserves convexity.** Three ways to prove it: epigraph intersection, case-splitting, or the max-of-sums lemma. The epigraph proof uses the key equivalence $\max(a, b) \leq t \iff a \leq t$ and $b \leq t$.
4. **Intersection preserves set convexity** (but union does not). Each constraint in an optimization problem is a convex set; their intersection (the feasible set) is also convex.